In Keun Kim

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EDUCATION

Columbia University — New York, NY

M.S. in Computer Science, Advanced Master's Research (AMR) Specialization — Expected May 2027

Faculty-nominated for extended graduate research; advised by Prof. Baishakhi Ray.

Seoul National University — Seoul, South Korea

B.S. in Industrial Engineering, Minor in Computer Science — Mar 2015–Aug 2024

Completed while working full-time as a software engineer.

Alternative military service: Oct 2019–Aug 2022 (34 months).

RESEARCH INTERESTS

Programming languages and program analysis, software testing and verification, and cybersecurity for reliable and secure AI-generated software.

PUBLICATIONS AND MANUSCRIPTS

[1] Elaine Ang, **In Keun Kim**, Sam Weldon, Kevin Durand, Kostis Kaffes, and Eugene Wu. "BranchBench: Aligning Database Branching with Agentic Demands." arXiv:2604.17180, 2026. [Paper] [Code]

[2] Hailie Mitchell, **In Keun Kim**, Song Li, Alex Mathai, Junfeng Yang, Baishakhi Ray, and Zhuo Zhang. "The Illusion of Correctness: Measuring Semantic Divergence in Agent-Generated Security Patches." Submitted, 2026.

[3] **In Keun Kim**, Hyesung Son, Sangyoung Oh, and Youngeun Cho. "Analytical Cost Modeling of Multi-Token Prediction and Head-Count Optimization for LLM Inference." Manuscript, 2026.

RESEARCH EXPERIENCE

Cybersecurity Agent Topology and Orchestration

Columbia University · Advised by Prof. Baishakhi Ray — *Sep 2025–Present*

- Designed and implemented a hierarchical multi-agent system to test whether role-based decomposition improves automated security repair, combining a shared planner, specialized managers, bounded workers, and host-enforced permissions.
- Evaluated flat and hierarchical configurations on held-out CVEs and analyzed patch quality, cost, tool use, and provenance; preliminary results show that hierarchy alone does not compensate for weak reasoning or validation workers.

The Illusion of Correctness: Agent-Generated Security Patches

Columbia University · with Prof. Junfeng Yang, Prof. Baishakhi Ray, and Prof. Zhuo Zhang — *Feb 2026–Jun 2026*

- Led a statistical analysis of whether patch size predicted unintended behavioral changes in agent-generated security repairs, using CVE-matched models across 474 patches from 79 CVEs.
- Found that larger patches were associated with both higher odds and greater diversity of semantic divergence; the study's differential fuzzing detected divergences in 6× more patches than augmented regression testing.

BranchBench: Database Branching for Agentic Exploration

Columbia University · with Prof. Kostis Kaffes and Prof. Eugene Wu *Feb 2026–Apr 2026*

- Co-developed parameterized macrobenchmarks for five agentic exploration workloads and microbenchmarks isolating database branch-lifecycle costs.
- Evaluated Neon, DoltgreSQL, TigerData, Xata, and PostgreSQL baselines, exposing a 25–1,500× branch-latency versus 5–4,000× read-latency trade-off.

Analytical Cost Modeling of Multi-Token Prediction and Head-Count Optimization

Independent collaboration · with Prof. Youngeun Cho (Hanyang University) *Jan 2026–Jun 2026*

- Formulated and proved an exact finite-length expected-cost model for MTP and an O(1)-computable safe bound for exact head-count search.
- Validated the selected configuration with Llama2-7B/70B simulations and an Alpamayo-R1 GPU implementation, yielding an 83.6% average modeled cost reduction and a 1.74× measured speedup over the single-head baseline.

TEACHING AND MENTORING EXPERIENCE

COMS E6998-019: Designing Production Agentic Systems, Teaching Assistant — Fall 2026

Research Mentor, two high school students on my research project — Apr 2026–Present

PROFESSIONAL EXPERIENCE

Machine Learning Engineer, Viva Republica (Toss) — Jul 2024–Feb 2025

- Built a DeepFM inference platform handling 6,000+ RPS with p95 latency below 70 ms and cut model-release time from 6 hours to 30 minutes through MLflow packaging, zero-downtime deployment, and automated data-integrity tests.

Server Engineer, Viva Republica (Toss) — Sep 2022–Jun 2024

- Built a fraud-scoring pipeline processing over 1 TB of features, with p95 latency of 200 ms and p99.9 latency of 600 ms.

Python Developer, Viva Republica (Toss) — Oct 2019–Aug 2022

- Prototyped Toss's first fraud-detection inference server using LightGBM, PyTorch, and Kafka.

Frontend Developer, XINICS — Jul 2018–Oct 2019

- Built React extensions for Canvas LMS and automated database and NAS provisioning across 30 Korean universities with zero installation errors.

Software Engineer, AKA Intelligence — Sep 2017–Feb 2018

- Built a Django manufacturing execution system and customized the embedded Android OS with an English vocabulary-learning feature.

HONORS AND SCHOLARSHIPS

Samsung Convergence Software Course (SCSC) Scholarship — Spring 2018

Merit-Based Scholarships (three awards), Seoul National University — 2015–2016

TECHNICAL SKILLS

Languages: Python, Kotlin, Java, Javascript, Typescript, Scala, Rust, C

Machine Learning & Serving: PyTorch, TensorFlow, CatBoost, LightGBM, MLflow, ONNX, NVIDIA Triton

Systems & Data: Kafka, Airflow, Kubernetes, Docker, PostgreSQL, MySQL, HBase, Spark